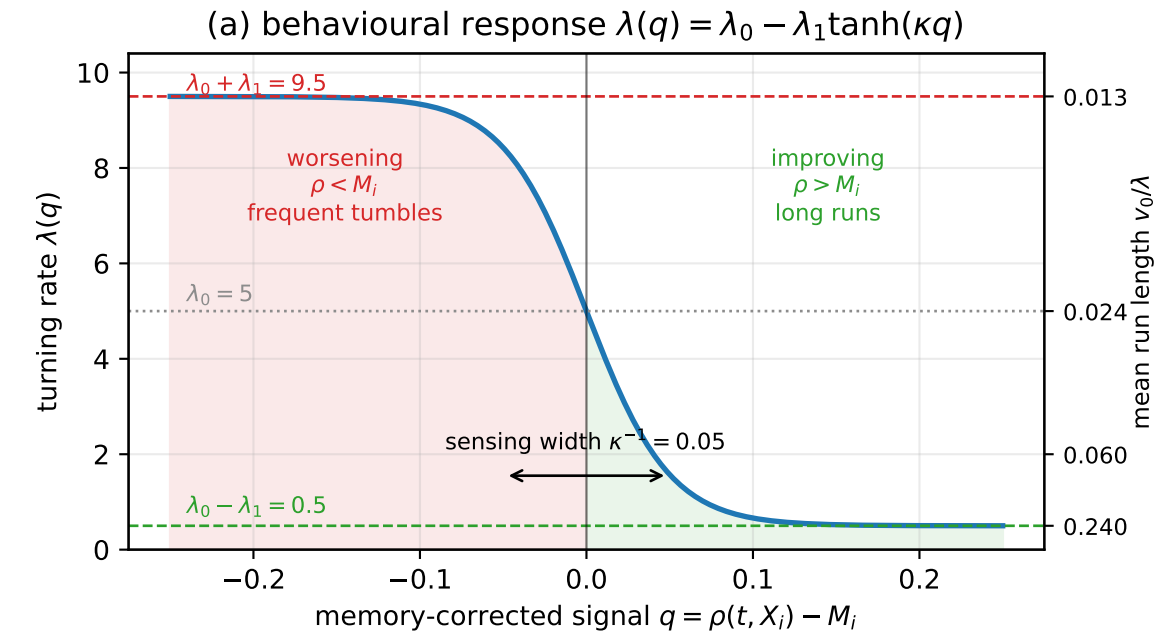
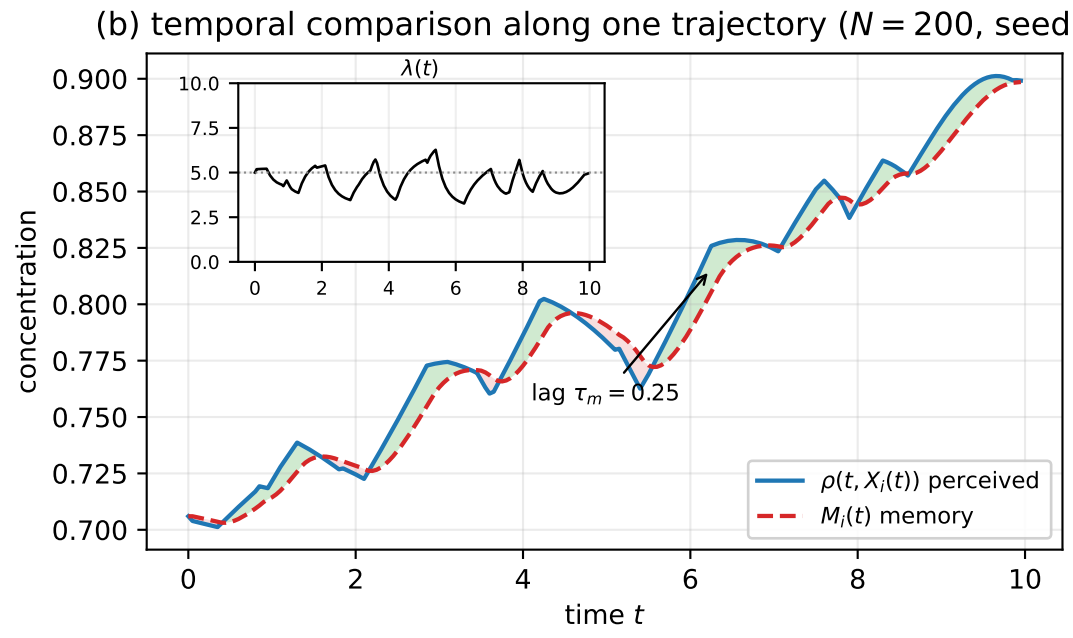
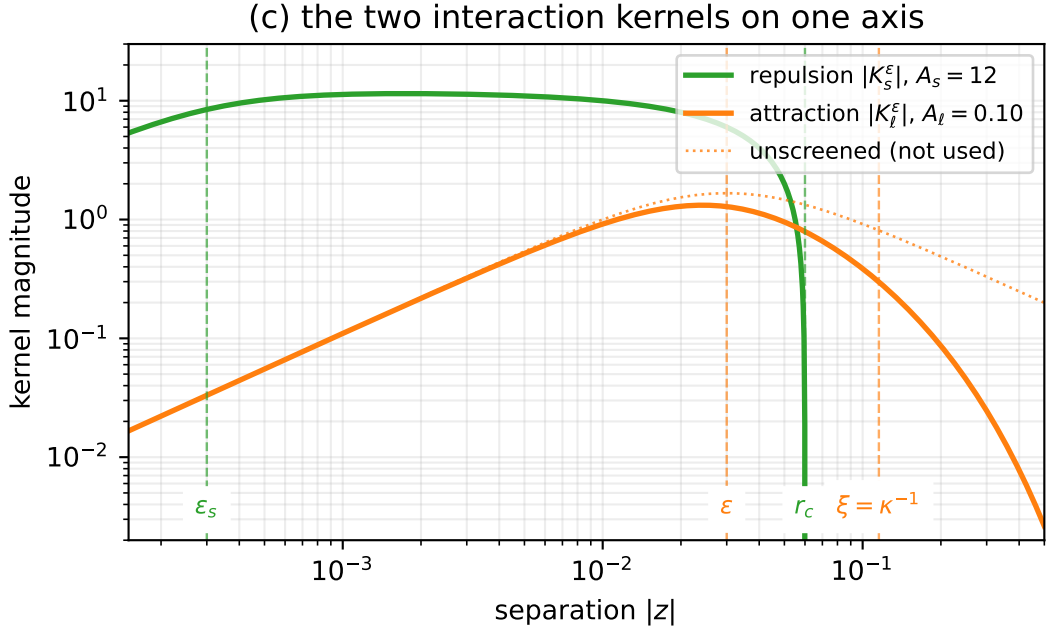
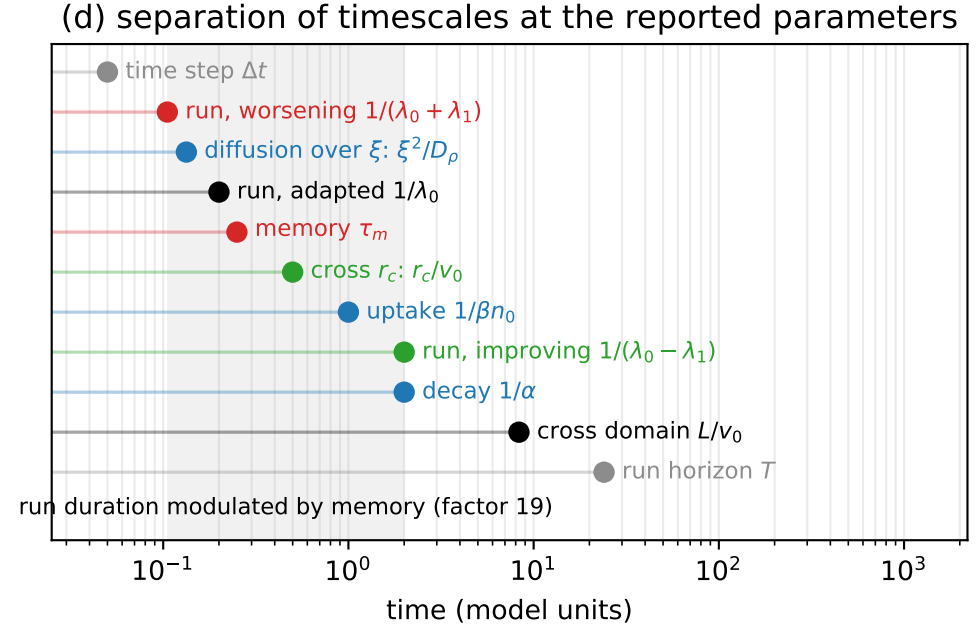

(a) behavioural response $\lambda(q) = \lambda_0 - \lambda_1 \tanh(\kappa q)$
$\lambda_0 + \lambda_1 = 9.5$
worsening
$\rho < M_i$
frequent tumbles
improving
$\rho > M_i$
long runs
$\lambda_0 = 5$
sensing width $\kappa^{-1} = 0.05$
$\lambda_0 - \lambda_1 = 0.5$
turning rate $\lambda(q)$
mean run length v_0/λ
0.013
0.024
0.060
0.240
memory-corrected signal $q = \rho(t, X_i) - M_i$


(b) temporal comparison along one trajectory ($N = 200$, seed 7)
$\lambda(t)$
lag $\tau_m = 0.25$
$\rho(t, X_i(t))$ perceived
$M_i(t)$ memory
concentration
time t


(c) the two interaction kernels on one axis
repulsion $|K_s^\varepsilon|$, $A_s = 12$
attraction $|K_\ell^\varepsilon|$, $A_\ell = 0.10$
unscreened (not used)
ε_s
ε
r_c
$\xi = \kappa^{-1}$
kernel magnitude
separation $|z|$


(d) separation of timescales at the reported parameters
time step Δt
run, worsening $1/(\lambda_0 + \lambda_1)$
diffusion over ξ: ξ^2/D_ρ
run, adapted $1/\lambda_0$
memory τ_m
cross r_c: r_c/v_0
uptake $1/\beta n_0$
run, improving $1/(\lambda_0 - \lambda_1)$
decay $1/\alpha$
cross domain L/v_0
run horizon T
run duration modulated by memory (factor 19)
time (model units)